Padraig Lavery

Software Engineer – Reliability, Automation & Infrastructure as Code

Belfast, Northern Ireland | LinkedIn | GitHub

PROFILE

Software engineer with 4+ years keeping high-volume, security-sensitive banking platforms reliable on AWS and Kubernetes. Uses software to remove operational toil: automating manual jobs, fixing root causes rather than symptoms, and making batch systems fail safely. Has led cross-team engineering work and owns fixes from code through to production release.

EXPERIENCE

Developer | Reward, Belfast | June 2022 – Present

Reliability engineering

- Eliminated recurring daily out-of-memory crashes in a high-volume transaction ingestion service by loading caches once per batch instead of per file; diagnosed and mitigated further memory-exhaustion failures in ETL jobs.
- Redesigned a month-end transactional invoicing batch with per-partner error isolation, explicit timeouts and safe retry/reconciliation, so failures on the highest-volume partners no longer abort the whole run.
- Resolved a live customer-facing outage caused by malformed load-balancer group naming in a Kubernetes/Istio ingress setup, and fixed the Helm templates so it could not recur.

Reducing toil

- Automated scheduled file processing with EventBridge, Lambda and Fargate, removing manual triggers.
- Wrote runbooks and SOPs so the production services team resolves recurring issues without engineering escalation.
- Audited 100+ S3 buckets, applying lifecycle policies and removing obsolete resources to cut storage cost and compliance risk.

Security-sensitive systems

- Remediated critical Snyk-flagged vulnerabilities for PCI DSS compliance; wrote the AWS SDK v2-to-v3 migration guide and planned testing for Node 22 runtime upgrades across Lambdas, Fargate services and microservices.
- Hardened authentication and account services by removing stack traces and SQL details from error responses and standardising input validation.
- Built audit logging of staff access to customer accounts across UK and Middle East bank portals.

Delivery

- Led a cross-team initiative for automated API testing and stabilised GitHub Actions CI, including fixing a GitHub App token authentication issue that had blocked test runs.
- Mentored four new starters; run the team's retrospectives.

Intern Software Engineer | Flexera, Belfast | May 2020 – April 2021

- Built a cloud-cost API with Go, Docker and Kubernetes.

SKILLS

- **Reliability and operations:** incident response, root cause analysis, runbooks/SOPs, safe retries and timeouts, batch resilience
- **Infrastructure as code and cloud:** Terraform, Helm, Kubernetes, Istio, Docker, AWS (Lambda, Fargate, S3, EventBridge), GitHub Actions
- **Languages:** JavaScript (Node.js), Python, C#, Go, SQL
- **Data:** PostgreSQL, MongoDB
- **Security:** Snyk, OWASP, PCI DSS remediation

PROJECTS

Incident Copilot (in progress) | Python, FastAPI, PostgreSQL + pgvector, Ollama, Docker, GitHub Actions

- Incident-triage service that matches an incident description to runbook content and suggests likely causes and next steps.

EDUCATION AND CERTIFICATIONS

- BSc (Hons) Computer Science, 2:1 – Ulster University, Belfast, June 2022

- AWS Certified Developer – Associate; AWS Certified Cloud Practitioner
- AWS Certified AI Practitioner – preparing, exam November 2026